

When Adapters Don't Add Up: Testing Parameter-Space Compositionality of Language and Genre

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Abstract

The goal of this paper is to test whether a language model's competence in a language and its competence in a genre, each acquired independently through a small low-rank adapter, can be recombined to model their intersection by direct summation in weight space, without any joint training signal. Using facebook/xglm-564M and two OPUS-derived corpora spanning legal/administrative and literary registers, independent LoRA modules are trained for Finnish and for legal-register text, then compose them zero-shot and, separately, after a small amount of real target-domain supervision.

1 Introduction

Language models pretrained on web-scale corpora achieve strong performance on the languages, time periods, and genres that dominate their training data, but degrade sharply when confronted with text that lies at the intersection of several under-represented factors simultaneously. A nineteenth-century Maltese newspaper article and a contemporary Maltese pharmaceutical leaflet share a language but differ in register, genre, and diachronic layer; a model that has seen substantial Maltese legal text and substantial nineteenth-century French prose has, in principle, been exposed to much of the linguistic substance needed to model either domain, yet has likely seen almost none of their conjunction. This raises a question that is as much about the geometry of learned representations as it is about data availability: if a language model has separately acquired competence along individual factors of variation, say language, period, genre, can that competence be recombined to model a domain defined by their intersection, even when the intersection itself is severely data-scarce or entirely unobserved during adaptation?

This question is here investigated empirically for two factors of variation, language and genre, using two independently trained adaptation modules combined at inference time and, subsequently, with a small amount of supervised adaptation. The central research question, following the framing above, is whether linguistic variation of this kind is compositional in the parameter space of a language model: whether the modification a model undergoes to acquire genre competence and the modification it undergoes to acquire language competence can be linearly summed to yield a model competent in both simultaneously, without any joint training signal. This has been operationalized by composing two independently trained low-rank adaptation (LoRA) modules through summation of their weight deltas, which lets us test the parameter-space compositionality question in its most literal form.

The following sections describe the backbone model and modular adaptation architecture, and justifies the choice of parameter-space composition via low-rank adaptation as the primary experimental condition. Then, the target domain and data splits are described, as well as the evaluation protocol, and report results, including a negative result for naive parameter-space composition discussed above.

2 Backbone and Modular Architecture

Backbone selection Facebook/xglm-564M ((Lin et al., 2021)) is adopted as the backbone language model. EuroLLM-1.7B (Martins et al., 2024) was also a candidate, given its stronger and more contemporary European-language coverage, but found it computationally impractical within the available compute budget.

Three considerations motivate the final choice over alternatives. First, unlike backbones whose multilingual competence is a secondary property of a training mixture dominated by one or two

high-resource languages, XGLM was explicitly trained on a corpus balanced across a broad typological and genealogical range of languages, giving more defensible and comparable coverage across the language set available in our target corpora (Helsinki-NLP/opus_dgt, Helsinki-NLP/opus_books) than backbones whose multilingual capacity is comparatively incidental. This matters directly for our design: an uneven backbone would confound the compositionality question asked in this context with simple differences in per-language pretraining exposure. Second, XGLM is a decoder-only, causally trained model, which admits a standard, unambiguous evaluation metric (token-level perplexity under next-token prediction) without recourse to the pseudo-log-likelihood approximations required for masked encoders (Salazar et al., 2020). Finally, at 564M parameters, XGLM keeps both independent module training and the composed-model evaluation pipeline computationally tractable within the compute budget available.

Modular adaptation architecture Low-rank adaptation (Hu et al., 2021) is adopted as the adaptation module, in preference to the bottleneck-adaptor architecture of Pfeiffer et al. (2020) and its cross-lingual extension: a LoRA module parameterizes a factor-specific update to a subset of the backbone’s weight matrices as a low-rank decomposition applied additively to the frozen pre-trained weight W . Because this update is expressed directly in the parameter space of the model, two independently trained LoRA modules can be composed by simple summation of their deltas, $W_{comp} = W_l + W_g$, prior to merging into the backbone. This operation is the most direct available instantiation of the question of whether linguistic variation is compositional in the parameter space of a language model, and it is treated as the primary composition operator for that reason: it is the operationalization that corresponds most literally to the research question as posed.

3 Data

Two parallel corpora distributed via the OPUS collection and made available through the Hugging Face Hub have been used: Helsinki-NLP/opus_dgt, a corpus of legal and administrative text drawn from the EU’s Directorate-General for Translation and covering 25 languages, and Helsinki-NLP/opus_books, a corpus of literary

text covering 16 languages (Tiedemann, 2012).

This pairing determines the structure of the module training data: the genre module for our target genre is trained on DGT restricted to all languages other than the target language l , while the language module for l is trained on Books restricted to l . This design has a consequence worth stating, as it bears on how the resulting modules should be interpreted. Because Books is itself a single-genre corpus, the language module is not trained on a genre-neutral sample of l ; it is trained on l as instantiated in literary prose specifically. There is no way, within the two datasets made available for this assignment, to construct a training signal for l that is not simultaneously a training signal for some particular genre, since every sentence in either corpus carries both a language label and a genre label by construction. What is called the "language module" is therefore more precisely a module for l -in-literary-register, not language competence disentangled from genre. This has been treated as an inherent property of the two-dataset design available, a resource constraint rather than an implementation choice.

4 Methodology

Target domain selection Selection: l = Finnish and g = legal/administrative text (OPUS DGT) as the target intersection domain. Finnish is a useful test case for the compositionality question beyond data availability: as a Uralic, agglutinative language, it is typologically distant from the Indo-European, predominantly fusional languages that dominate both source corpora, so a positive compositionality result for Finnish would be harder to attribute to superficial cross-lingual similarity than the same result for, say, a Romance or Germanic target language. The target-domain corpus is split into training, validation, and test portions (data/target_domain_fi_dgt/): 52,509 sentences (999,994 whitespace tokens) for training, 2,630 sentences (50k tokens) for validation, and 2,618 sentences (50k tokens) held out for test. All reported metrics use the test portion exclusively; the validation portion is reserved for any hyperparameter decisions made during the lightweight supervised adaptation phase.

Module training Both the language and genre modules are implemented as LoRA adapters (Hu et al., 2021) on the frozen XGLM backbone, using identical hyperparameters so that the result-

ing weight deltas are directly combinable in a shared parameter space: rank $r = 8$, $\gamma = 16$, dropout 0.05, applied to the query, key, value, and output projection matrices, trained with learning rate $2e-4$, batch size 4, for 600 steps. Each adapter adds approximately 1.57M trainable parameters (0.28% of the backbone). These values follow standard defaults from the original LoRA formulation rather than a tuned configuration, since the available compute did not support a hyperparameter search; what matters is that both modules share an identical configuration, which is a precondition for their weight deltas being directly summable in a shared parameter space. Each module is independently trained with a standard causal language-modeling (next-token prediction) objective via the Hugging Face ‘Trainer’, using a fixed, deliberately lightweight compute budget (a modest number of optimization steps, small batch size, short linear warmup). Because LoRA’s up-projection (B) matrix is zero-initialized, the adapter contributes nothing to the forward pass before any training occurs; this is exploited to obtain a "pre-adaptation" reference point for each module directly from the same evaluation call used at step 0, without needing to load or evaluate a separate unadapted copy of the model. "Independently trained" means each module starts from its own fresh copy of the frozen backbone and freshly initialized parameters, sees data disjoint from the other module along both factors, and is saved separately from the backbone so it can later be loaded, recombined, or discarded on its own.

Zero-shot composition The first composition setting asks whether the two independently trained, factor-specific adapters can be combined at inference time, with no additional training and no target-domain data of any kind, to better model their intersection domain than either adapter, or no adaptation at all, would alone. The composition operator used is direct addition in weight space: the composed update is a weighted sum of the two LoRA deltas, with equal weighting between the two modules by default. This is implemented via PEFT’s adapter-combination utility in its linear/additive mode, which produces a single new adapter representing the combined update without altering either of the two original, independently-trained adapters. This operator was chosen as the simplest and most standard form of "task arithmetic" available for LoRA-style mod-

ules: it introduces no new parameters, requires no additional data or training whatsoever, and makes no a priori assumption about which of the two factors should dominate the combination. One structural property of this specific operator is worth stating: weight-space addition is commutative, so the order in which the two adapters are supplied to the combination step cannot affect the resulting model, by construction.

Four conditions are evaluated under this setting, all without any gradient updates: i) the untouched backbone (no adaptation at all, serving as the floor reference); ii) the genre module alone; iii) the language module alone; iv) and the two modules composed as described above. Comparing all four against one another and against the floor reference is what isolates the individual zero-shot transfer contribution of each factor module from whatever additional effect, positive or negative, composing them together produces. An SVD-based combination that compresses the summed update back to a fixed target rank was considered and rejected, since truncating to the top singular directions would discard part of the summed update and confound any resulting effect with the truncation itself, rather than isolating composition as such.

Lightweight supervised adaptation The second composition setting asks a different question: given a small amount of proper target-domain data ($\leq 1,000,000$ -token training split produced during data preparation) does starting from the composed modules offer any advantage over adapting a model to the target domain from scratch?

Concretely, the composed weight delta from the zero-shot setting is permanently merged into the backbone’s own weights, producing a single frozen model in which both modules’ combined contribution is set into every forward pass rather than represented as a separate, removable adapter. A new LoRA adapter, freshly initialized, using the identical configuration as the two genre/language modules, is then trained on top of this frozen composed backbone, using the target-domain training split. This keeps the adaptation step itself lightweight (a small number of newly trained parameters, the same training budget used for module training) while allowing the model to draw on real, if scarce, in-domain supervision.

Two conditions are compared under identical training and evaluation settings: the adapter trained on top of the composed backbone, and

an otherwise identical adapter trained on top of the plain, unmodified backbone, using the same target-domain training data. The second condition is the critical non-modular control: it isolates whatever effect is attributable specifically to having pre-composed the two factor modules, as distinct from any effect that simply comes from adapting to the same amount of target-domain data regardless of starting point. A third, more heavy-weight condition (full-parameter fine-tuning of the backbone on the same target-domain data, with no LoRA adapter and no modules at all) was considered as an additional baseline, but was not run. The backbone’s large tied embedding/output matrix (a consequence of its large multilingual vocabulary) makes full-parameter optimizer state a substantial memory cost, which wouldn’t have been feasible in this context.

Evaluation protocol

Metric Every condition in both settings is scored with the same intrinsic language-modeling metric: per-sequence average negative log-likelihood (equivalently, perplexity, reported as its exponential for interpretability). This metric was chosen because it requires no task-specific labels, given that, virtually, no downstream annotations exist for this low-resource intersection domain, and because computing it per example, rather than only as a single corpus-level aggregate, is what makes a paired statistical comparison between conditions possible in the first place.

Test-set and consistency All conditions, in both the zero-shot and the supervised-adaptation setting, are scored on the same held-out target-domain test split, in the same fixed sentence order, with the same tokenization and truncation settings used throughout the project. This is what allows per-example scores to be compared directly, sentence-for-sentence, across every condition rather than only as aggregate statistics.

Baseline The full set of conditions across both settings is designed as one coherent comparison. The untouched backbone and the backbone adapted directly on the target-domain data (no modules, in the zero-shot and supervised settings respectively) serve as the two non-modular reference points against which any benefit of composition must be judged in each setting. The single-factor modules (genre-only, language-only) isolate

each factor’s individual zero-shot transfer to the target domain.

Statistical testing Because the same fixed set of test sentences is scored under every condition, comparisons between any two conditions are treated as paired. A paired Wilcoxon signed-rank test on the per-example log-likelihood differences is used rather than a parametric paired t-test, since sentence-level negative log-likelihood is not expected to be normally distributed (sentence lengths and lexical difficulty vary substantially within the legal-register test set). Per-sequence average negative log-likelihood (natural log) is computed manually (not ‘Trainer.evaluate’, which only returns a corpus-level mean).

5 Results

Table 1 summarizes both composition settings on the target-domain test set. In the zero-shot setting, the genre module alone yields a small improvement over the untouched backbone (32.05 vs. 33.73 perplexity), while the language module alone is worse than the backbone (43.33), and composing the two (simple weight-space addition of their LoRA deltas) is worse than every other zero-shot condition, including doing nothing at all (47.48). Once a small amount of real target-domain supervision is introduced, both adapted conditions improve sharply over every zero-shot condition, landing within a narrow band of each other (perplexity 20.02–20.23): adapting the plain backbone directly on the target-domain training split performs marginally better than adapting the same architecture on top of the pre-composed modules. Whether either gap - composed vs. backbone in the zero-shot setting, or composed-adapted vs. backbone-adapted in the supervised setting - is statistically meaningful is addressed in the next section.

Every condition across both settings was scored on the same 2,618 target-domain test sentences, in the same fixed order, with identical tokenization and loss computation. This matters because it is assumed that sentence difficulty varies substantially in this test set, and that variance is largely shared across conditions. An unpaired test (e.g. comparing the two raw loss distributions as if they came from unrelated samples) would treat that shared variance as noise and dilute the comparison’s sensitivity. A paired test instead looks only at the within-sentence difference, cancelling out

Condition	PPL ↓
Backbone	33.73
+ Genre module (zero-shot)	32.05
+ Language module (zero-shot)	43.33
+ Both, composed (zero-shot)	47.48
Backbone, adapted	20.02
Composed, adapted	20.23

Table 1: Perplexity on the Finnish-DGT target-domain test set. First block: zero-shot (no gradient updates). Second block: adapted on the $\leq 1\text{M}$ -token target-domain training split. Lower is better.

sentence-level difficulty and leaving just the effect of the condition.

Realistically, the natural paired parametric test would be a paired t-test on the per-sentence NLL differences, which assumes those differences are normally distributed. That assumption is not a good fit here: sentence-level NLL is bounded below by zero and right-skewed (most sentences are relatively easy; a tail of long or lexically unusual sentences produces disproportionately large losses), and one model occasionally handles a particular sentence far worse than another (e.g. an out-of-domain token sequence), producing large individual differences that would unduly influence a mean-based test. The Wilcoxon signed-rank test only requires the differences be roughly symmetric around their median under the null hypothesis of no effect and operates on the ranks of the differences rather than their raw magnitudes, making it less sensitive to those outliers. It is the standard nonparametric analogue of the paired t-test and a common choice for paired NLP evaluation metrics for exactly this reason.

Because six comparisons are drawn from overlapping conditions, the Holm-Bonferroni correction has been applied across all tests to control the family-wise error rate at $\alpha = 0.05$, rather than interpreting each p-value in isolation.

6 Discussion

The central finding is a clear negative result for the primary research question. Simple, equally-weighted addition of two independently trained LoRA deltas does not produce a model competent in both the target language and the target genre at once: it produces a model worse than either factor module alone, and worse than doing nothing at all. Under the specific conditions tested here, lin-

guistic variation of this kind is not compositional in the model’s parameter space.

Two features of the experiment design help explain why, and both point to the same underlying issue: neither single-factor module is a clean, uncontaminated signal for the factor it is supposed to represent.

The language module is confounded with genre by construction. As noted in the Data section, the only training signal available for the target language comes from the literary portion of our corpora, so this module is better described as a module for the target language in literary register specifically, not the target language independent of register. Applied to a legal-register test domain, it does not simply supply "Finnish": it actively pulls the model toward literary patterns that conflict with the legal register being evaluated. This offers a direct explanation for why the language module alone performs worse than the untouched backbone (43.33 vs. 33.73 perplexity): its contribution is not genre-neutral, and here that mismatch outweighs whatever benefit its language-specific signal provides.

The genre module is confounded with language coverage. Because the backbone’s pretraining data includes only a minority of the languages present in the genre module’s training corpus, much of that module’s training signal is not purely teaching legal-register conventions in languages the model already knows: it is partly teaching some of those languages themselves, from a small adapter budget never designed for that task. That the genre module still yields a small zero-shot improvement (32.05 vs. 33.73) suggests some real register-specific signal survives this confound, likely carried by the languages that do overlap with the backbone’s pretraining data, but its contribution should not be read as a clean measurement of genre transfer on its own.

Combining two confounded, individually imperfect signals by direct addition gives the model no way to resolve or arbitrate between them. Each delta was optimized on its own, with no signal during training that it would later be added to another. Adding weight-space updates together implicitly assumes the two updates point in compatible, non-conflicting directions within the same weight matrices; nothing about how the modules were trained enforces this, and the result (47.48 perplexity, worse than either input alone) is con-

Comparison	Median Δ NLL	Win rate	p (Holm)	Sig.
Composed vs. backbone	+0.330	10.7%	$< 10^{-300}$	Yes
Composed vs. genre-only	+0.377	4.5%	$< 10^{-300}$	Yes
Composed vs. language-only	+0.114	27.4%	1.15×10^{-109}	Yes
Composed-adapted vs. backbone-adapted	+0.008	42.3%	4.58×10^{-21}	Yes
Composed-adapted vs. composed	-0.701	99.8%	$< 10^{-300}$	Yes
Backbone-adapted vs. backbone	-0.390	98.8%	$< 10^{-300}$	Yes

Table 2: Paired Wilcoxon signed-rank tests on per-sentence NLL, $n = 2,618$ test sentences, Holm-Bonferroni corrected across all six tests ($\alpha = 0.05$). Median Δ NLL and win rate are computed as (first condition $-$ /vs. second); a positive median Δ NLL means the first-named condition is typically *worse*. All raw p -values underflow to machine zero except where shown; exact values in `results/statistical_tests.csv`.

sistent with the two updates interfering with each other rather than combining constructively. This pattern is not unique to this setting: related work (Yadav et al., 2023) on merging independently fine-tuned models has documented similar interference when weight-space deltas are added without any mechanism to resolve conflicting updates, and has proposed fixes such as pruning small components or resolving sign disagreements before merging. Plain, unmodified addition is used deliberately, as the most direct test of the research question as posed; whether a more sophisticated way of combining the deltas would close some of this gap should be a next step in this research field, discussed below.

The picture changes substantially once real target-domain supervision is introduced. Both the composed-then-adapted and the plain-backbone-then-adapted conditions land close to each other (perplexity 20.02–20.23), a sharp contrast with their zero-shot counterparts (47.48 vs. 33.73). With only about a million tokens of real target-domain data, whatever damage the zero-shot composition caused is almost entirely undone. The remaining gap between the two adapted conditions is statistically distinguishable after correcting for multiple comparisons, but its practical size tells a more equivocal story: composed-adapted beats backbone-adapted on only 42.3% of individual test sentences, close to a coin flip, and the median per-sentence difference is two orders of magnitude smaller than the zero-shot effects above. This is the familiar large-sample pattern in which statistical and practical significance pull apart — with 2,618 paired sentences, even a small, fairly consistent tilt toward one condition is detectable, but it does not represent the kind of clear, sentence-by-sentence dominance seen in the zero-shot setting. Pre-composing the two modules does not

meaningfully help over adapting the plain backbone from scratch once real supervision is available, and there is weak evidence that it is very slightly worse. The two adaptation checks both show near-total win rates (98.8–99.8%), consistent with the large improvements already visible in the raw perplexity numbers, and serve as a useful confirmation that the evaluation pipeline is behaving as expected.

It is necessary to be careful about how far these results generalize. This is only one way of combining the modules (unweighted addition), one backbone, one language-genre pair, and two datasets whose construction makes fully separating language from genre impossible. A negative result under these specific conditions is evidence against this particular way of operationalizing parameter-space compositionality; it is not evidence that no such operationalization could succeed.

7 Future work

Several follow-ups would help separate the confounds discussed above from the compositionality question itself. A mismatched-module control, so composing the genre module with a language module trained on a different language, or vice versa, would test whether the interference that has been observed is specific to this factor pairing or a generic property of summing any two independently trained deltas. Weighted composition, giving one module a smaller coefficient than the other rather than assuming equal contribution, would test whether the negative result is an artifact of the unweighted operator rather than of composition itself. The sequential, representation-space composition as a comparison condition remains a natural next step, and would let the order-sensitivity question be asked in a setting where order can actually

matter. Finally, repeating this design with a genre-neutral source of language-only training data, and with a backbone whose pretraining coverage more fully overlaps the languages used to train the genre module, would help isolate parameter-space compositionality from the specific data and coverage constraints of this study.

8 Conclusion

The goal of this experiment setting is to test whether a language model’s competence in a language and its competence in a genre, each acquired independently through a small low-rank adapter, can be recombined by direct summation in weight space to model their intersection, without any joint training signal. Under the specific conditions of this study, the answer is no: composing the two adapters is worse than using either alone or using neither, and the modules themselves carry confounds (see a language module entangled with a specific register, a genre module partly relearning under-covered languages) that complicate a clean reading of the result. A small amount of true target-domain supervision resolves most of this gap, though it does not clearly favor the pre-composed starting point over adapting the plain backbone directly. This should be considered as a preliminary but concrete negative result for *naive* parameter-space composition of these two factors, consistent with, though not conclusive proof of, the possibility that the intersection domain in question carries structure not reducible to a simple sum of its factors.

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